Data communications (Lab 3): Phase Shift Keying (PSK)

Objective: To generate and analyze a Phase Shift Keying (PSK) signal using MATLAB by modulating a binary data sequence with a sinusoidal carrier signal.

Theory

Phase Shift Keying (PSK) is a digital modulation technique where the **phase of the carrier signal is changed according to the binary data**.

In **Binary PSK (BPSK)**:

- Binary **1 → phase = 0°**
- Binary **0 → phase = 180° (π radians)**

Instead of using 0 and 1 directly, we convert:

- 1 → +1
- 0 → −1

So, the modulated signal becomes:

$$s(t) = b(t). \sin(2\pi f t)$$

Where:

- $b(t)$ = bipolar data (+1 or −1)
- $\sin(2\pi f t)$ = carrier signal

  When $b(t). = -1$, the signal is inverted → phase shift of 180°

```
clear;
clc;

b = [0 1 0 1 1 1 0];     % Binary data input
n = length(b);           % Number of bits

t = 0:0.01:n;            % Time vector

% Convert binary (0,1) to bipolar (-1,+1)
for i = 1:n
    if (b(i) == 0)
        b_p(i) = -1;     % 0 → -1
    else
        b_p(i) = 1;      % 1 → +1
    end
end

% Generate waveform (expand each bit)
for j = 1:n
    bw(j*100:(j+1)*100) = b_p(j);
end
```

```
bw = bw(100:end);       % Adjust length

% Generate carrier signal
sint = sin(2*pi*t);

% Perform PSK modulation
st = bw .* sint;

% -------- Plotting -------- %

subplot(3,1,1)
plot(t,bw)
grid on
axis([0 n -2 +2])
title('Bipolar Data Signal')

subplot(3,1,2)
plot(t,sint)
grid on
axis([0 n -2 +2])
title('Carrier Signal')

subplot(3,1,3)
plot(t,st)
grid on
axis([0 n -2 +2])
title('PSK Modulated Signal')
```

Code Explanation

1. Initialization

```
clc;
clear;
close all;
```

- `clc` → Clears command window
- `clear` → Removes all variables
- `close all` → Closes all figure windows

2. Binary Input

```
b = [0 1 0 1 1 1 0];
```

- Input data sequence

3. Convert to Bipolar Signal

```
if (b(i) == 0)
    b_p(i) = -1;
else
    b_p(i) = 1;
end
```

Converts:

- $0 \rightarrow -1$
- $1 \rightarrow +1$

Required for PSK modulation

4. Generate Bit Waveform

```
bw(j*100:(j+1)*100) = b_p(j);
```

- Expands each bit into 100 samples
- Creates a rectangular waveform

5. Carrier Signal

```
sint = sin(2*pi*t);
```

- Generates sine wave carrier

6. PSK Modulatio

```
st = bw .* sint;
```

Multiplies bipolar signal with carrier

- $+1 \rightarrow$ normal sine wave
- $-1 \rightarrow$ inverted sine wave (phase shift 180°)

7. Plotting

- Subplot 1 → bipolar data
- Subplot 2 → carrier signal
- Subplot 3 → PSK signal

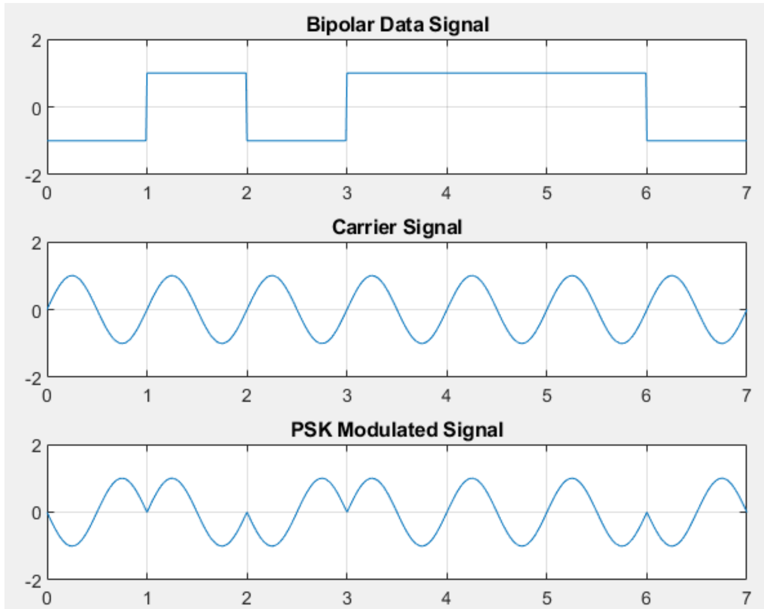
<u>Result</u>

The MATLAB program produces:

1. Bipolar digital signal (+1, −1)
2. Carrier sine wave
3. PSK modulated signal

Observations:

- When bit = 1 → normal sine wave
- When bit = 0 → inverted sine wave



Conclusion: The experiment successfully demonstrated Phase Shift Keying using MATLAB. The binary data was converted into a bipolar signal and used to modulate a carrier wave. The output showed correct phase shifts corresponding to the input bits, verifying the principle of PSK.

Assignment: Write a MATLAB program to generate an FSK signal for the binary sequence: b = [1 0 1 1 0 0 1]
Requirements:

- Use two frequencies:
  - ✓ f1=5 Hz (for bit 1)
  - ✓ f0=2 Hz (for bit 0)
- Generate:
  - ✓ Binary signal
  - ✓ FSK modulated signal
- Plot both signals